

## UCC504: SURFACE HYDROLOGY

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|----------|----------|----------|------------|
| <b>L</b> | <b>T</b> | <b>P</b> | <b>Cr</b>  |
| <b>3</b> | <b>1</b> | <b>0</b> | <b>3.5</b> |

**Course objective:** The overall objective of this course is to impart knowledge of the engineering aspects of the water science. The students will learn various components of hydrologic cycle, its processes, measurement and their engineering application. This course will also provide the exposure of software to students in solving the real-life problem.

**Introduction:** Hydrologic cycle, Scope and Applications.

**Probability and Statistics in hydrology:** Probability and random variables, Distribution function, Estimation of parameters, Correlation and Regression analysis using statistical softwares.

**Precipitation:** Measurement by rain gauge and other methods, estimation of missing data, consistency of records, optimum number of rain gauge station, mean precipitation, presentation and analysis of rainfall data.

**Abstractions from Precipitation:** Evaporation, factors affecting evaporation, measurement, infiltration, factors affecting infiltration, measurement, infiltration indices.

**Stream flow:** Run-off estimation, rainfall-runoff correlation, SCS-CN method, Stream flow measurement using various methods, flow duration curve, Flow mass curve, hydrographs, base flow separation, unit hydrographs and its application, synthetic unit hydrograph.

**Floods Frequency analysis:** Peak flood estimation, Flood frequency analysis, flood routing.

**Software application:** Watershed delineation and rainfall-runoff modeling using watershed model.

### Course Learning Outcomes:

Upon completion of this course, the students will be able to:

1. Perform analysis on different components of hydrological cycle.
2. Estimate the runoff generated from watershed using various methods/techniques.
3. Apply the principles of flood frequency analysis and flood routing to forecast floods.

### Text Books:

1. Subramanya, K., *Engineering Hydrology*, Tata McGraw-Hill Publication(2015)
2. Reddy, Jaya Rami., *A textbook of Hydrology*, University Science Press, New Delhi (2014)
3. Raghunath, H. M., *Hydrology: Principles, analysis and Design*, New Age International Publishers(2015)

### Reference Books:

1. Chow, V.T., Maidment, D.R., and Mays, L.W., *Applied Hydrology*, Mc-Graw-Hill International Editions, New York (2006)
2. Misstear, B., Banks, D., and Clark, L., *Water Wells and Boreholes*, John Wiley & Sons Ltd, UK(2013)
3. Shaw, E.M., Beven, K.J., Chappell, N.A., and Lamb, R. , *Hydrology in Practice*, Spon Press, New York(2014)
4. Viessman, W. and Lewis, G.L., *Introduction to Hydrology*, Prentice Hall of India Pvt Limited, New Delhi (2012)

**Evaluation Scheme:**

| <b>Sr. No.</b> | <b>Evaluation Elements</b>                                                          | <b>Weights (%)</b> |
|----------------|-------------------------------------------------------------------------------------|--------------------|
| 1.             | MST                                                                                 | 30                 |
| 2.             | EST                                                                                 | 40                 |
| 3.             | Sessionals ( May include<br>Assignments/Projects/Tutorials/Quizzes/Lab Evaluations) | 30                 |